Physics Insert: RUT, Quadratic Gravity, and Higher-Curvature Cosmology

Placement: Physics-only version - after the bridge-field action, FRW correspondence, or early-universe limits section.

Quadratic gravity provides a useful comparison class for Reverse Universe Theory because it attempts to connect ultraviolet gravitational behavior with inflationary cosmology through higher-curvature terms. In the recent paper "Ultraviolet Completion of the Big Bang in Quadratic Gravity," Liu, Quintin, and Afshordi describe a quantum quadratic-gravity inflationary scenario in which the ultraviolet theory is asymptotically free, one-loop running can drive slow-roll inflation toward the infrared, and general relativity re-emerges as an effective description after reheating.

The relevance for RUT is not that quadratic gravity replaces the RUT bridge. Rather, it provides a serious adjacent benchmark. Quadratic gravity modifies the Einstein-Hilbert action through curvature-squared operators. RUT, by contrast, introduces a bridge-sector and coupled field-duality structure intended to modify cosmological behavior while preserving a correspondence limit in which standard general relativity is recovered when the additional sectors are switched off or dynamically suppressed.

Comparison in action-language:

$$S_{QG} = \int d^4x \sqrt{-g} \left[(M_{Pl}^2/2)R + \alpha R^2 + \beta R_{\mu\nu}R^{\mu\nu} + \gamma C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \dots \right]$$

$$S_{RUT} = \int d^4x \sqrt{-g} \left[(M_{Pl}^2/2)R - 1/2\, K(B)(\nabla\phi)^2 - V(\phi,B) - 1/2(\nabla\chi)^2 - U(\chi) - 1/4\, Z(B)F^2 - 1/2\, \xi_Q B(\chi)(\nabla\theta \cdot \nabla\phi)^2 + L_m \right]$$

> The physics distinction is this: quadratic gravity primarily changes the curvature sector; RUT introduces a coupled bridge-and-field-duality sector whose observational role should be tested through late-time expansion, growth, lensing, and gravitational-sector consistency checks.

For the Physics version, the correct tone is comparative rather than triumphant. RUT should be described as a phenomenological and action-based framework that must be tested against LambdaCDM, quadratic gravity, scalar-tensor models, early dark energy, and other higher-curvature alternatives. Its advantage, if demonstrated, would not be that it discusses quantum gravity in general, but that it provides distinct observational signatures at late times: dynamic-omega behavior, possible ringing or log-periodic residuals in $H(z)$, modified growth behavior, and bridge-field consistency limits.

A reviewer-facing comparison should therefore ask five questions. First, does RUT reduce to GR in the correct limit? Second, does it avoid ghost, gradient, and strong-coupling pathologies in the perturbation sector? Third, does it improve fits to supernovae, BAO, RSD, CMB-derived priors, and lensing without excessive parameter freedom? Fourth, does it predict a tensor-sector signature distinguishable from quadratic gravity, including a possible prediction for r? Fifth, does it provide falsifiers that could kill the model if future data disagree?

The Liu-Quintin-Afshordi work predicts a minimum tensor-to-scalar ratio near $r = 0.01$ in order to avoid strong coupling. This gives RUT a practical target for future development: define whether the bridge sector predicts a different primordial tensor amplitude, a running tensor signature, or no independent early-universe tensor prediction at the present stage. Until that is derived, the responsible claim is that RUT is adjacent to, but not yet a replacement for, quadratic-gravity UV completion.

In technical positioning, RUT should be framed as a late-time and bridge-field cosmology with possible early-universe extensions. Quadratic gravity is a useful neighboring UV model. The two can be compared by their actions, correspondence limits, perturbation health checks, and observational predictions.

Source note: Based on Liu, Quintin, and Afshordi, "Ultraviolet Completion of the Big Bang in Quadratic Gravity" (arXiv:2510.18733), including the paper abstract describing asymptotic freedom, slow-roll inflation, GR emergence after reheating, and a minimum tensor-to-scalar ratio near r = 0.01.